Catch-up: the core of Lessons 1 and 2, and Lesson 3 up to Example 5

Module guide: Chapter 1, sections 1.4 to 1.4.4 (pages 21–28). There is no need to read it before the session.

BY THE END OF THIS LESSON YOU CAN

- Use the times table to multiply and divide, and say what × means ("groups of")
- Say what the top and the bottom of a fraction mean, and simplify a fraction
- Change an improper fraction into a mixed number, and back again
- Multiply a whole number by a fraction, and a fraction by a fraction
- Find a fraction OF an amount: divide by the bottom, then multiply by the top

HOW TO USE THESE NOTES

Every question in these notes has its full working **directly underneath it**, one small step at a time.

Cover the working with a piece of paper. Try the question yourself. Then slide the paper down **one step at a time** and compare each line with yours.

If your line is different, stop there. That step is the one to ask about.

Part 1 of 7: Times and divide, with the times table

The sign × means "groups of". Dividing asks "how many groups fit?"

PICTURE IT

The sign × is read "times". It means "groups of"

3 × 2 is read "3 times 2". It means **3 groups of 2**.

Count the dots: 2 + 2 + 2 = **6**.

So multiplying is a quick way of adding the same number again and again.



TOOL

The times table: keep it next to you all session

To find **6 × 7**: put a finger on row 6 and slide along to column 7. You land on **42**.

To **divide**, go backwards. For 42 ÷ 6, look along row 6 for 42. It sits under the 7, so 42 ÷ 6 = **7**.

Nobody has to know the tables by heart. Keep this page open in your notes.

×	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	**6**	**12**	**18**	**24**	**30**	**36**	**42**	**48**	**54**	**60**
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

PICTURE IT

Dividing asks: how many groups fit?

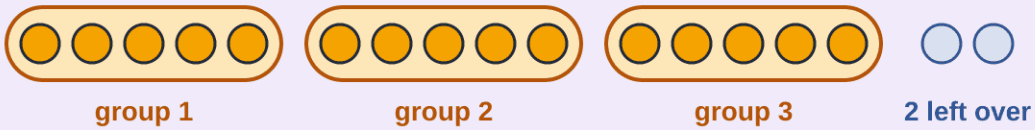
12 ÷ 4 asks: how many groups of 4 fit into 12?

Make groups of 4: three groups fit exactly. So 12 ÷ 4 = **3**.

Sometimes a few are left over. That is the **remainder**: 17 ÷ 5 = 3 remainder 2.



12 ÷ 4 = 3



17 ÷ 5 = 3 remainder 2

QUICK CHECK

Your go: 7 × 8, and 56 ÷ 7

Use the times table. Type the two answers.

7 × 8 = ?

56 ÷ 7 = ?

ANSWER

7 × 8 = 56

56 ÷ 7 = 8

Row 7, column 8 gives 56. Backwards: find 56 in row 7. It sits under the 8.

Part 2 of 7: What a fraction is, and how to simplify it

The top counts the slices. The bottom says how many equal slices make one whole

PICTURE IT

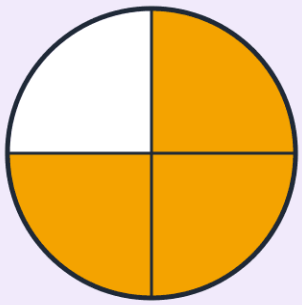
A pizza is cut into 4 equal slices. You eat 1

The whole pizza was cut into **4 equal slices**.

3 slices are left.

We write what is left as **3/4**.

We say "three quarters", or "three out of four".



$$\frac{3}{4}$$

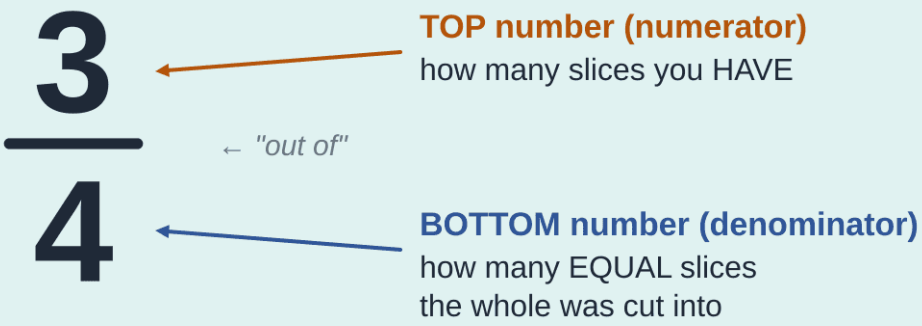
WORDS TO KNOW

Each number in a fraction has its own job

The **bottom** number says how many equal slices make one whole. Here it is 4, so the slices are quarters.

The **top** number counts how many slices you have. Here it is 3.

The module guide calls the top the **numerator** and the bottom the **denominator**. (**D**enominator: **D**own.)

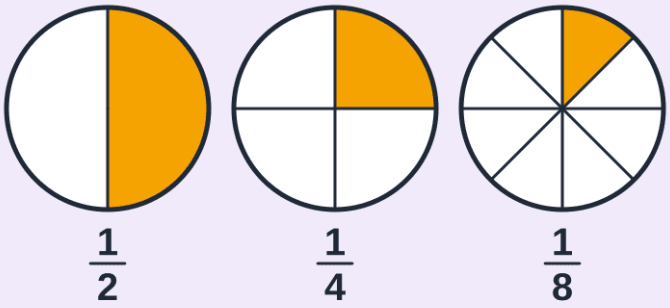


PICTURE IT

A bigger bottom number means SMALLER slices

Cut the same pizza into more slices, and each slice gets smaller.

So 1/8 is **smaller** than 1/4, even though 8 is bigger than 4.



QUICK CHECK

Your go: what fraction is orange?

Count the slices. Type the fraction like this: top/bottom



ANSWER

$\frac{5}{8}$

The pizza has 8 equal slices: that is the bottom. 5 of them are orange: that is the top.

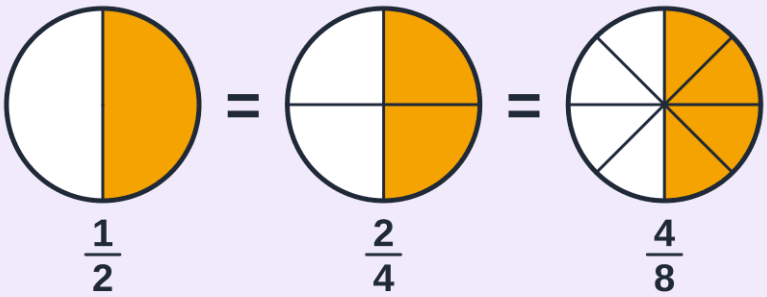
PICTURE IT

Half a pizza is half a pizza, however you cut it

All three pizzas have the **same amount** coloured.

Only the number of cuts is different.

So 1/2, 2/4 and 4/8 are the same amount.



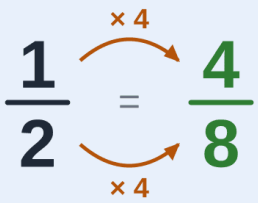
THE RULE

The golden rule of fractions

You may **multiply** the top and the bottom by the **same** number.

You may **divide** the top and the bottom by the **same** number.

The amount does not change. Only the numbers do.



WORDS TO KNOW

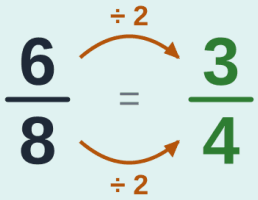
Simplifying: the same amount with smaller numbers

To **simplify**, divide the top and the bottom by a number that goes into **both**.

"Goes into" means it divides exactly, with nothing left over.

Keep going until **1 is the only number that goes into both**. Then the fraction is finished.

The exam says "reduced to the lowest term". It means the same thing.



Quick tests: does it go in exactly?

Goes in?	The test	Example
2	The last digit is even: 0, 2, 4, 6 or 8.	**36** ends in 6 ✓
3	Add the digits. The total is in the 3 times table.	**45**: 4 + 5 = 9 ✓
5	The last digit is 0 or 5.	**45** and **60** ✓
10	The last digit is 0.	**60** ✓

Worked example 1: simplify in small steps

Example 1

Simplify:

$$\frac{18}{30}$$

STEP 1

Find a number that goes into BOTH 18 and 30. Both are even, so 2 goes into both. Divide the top and the bottom by 2.

Why? Half of 18 is 9. Half of 30 is 15.

$$\frac{18}{30} = \frac{18 \div 2}{30 \div 2} = \frac{9}{15}$$

STEP 2

Look again. 9 and 15 are both in the 3 times table. Divide the top and the bottom by 3.

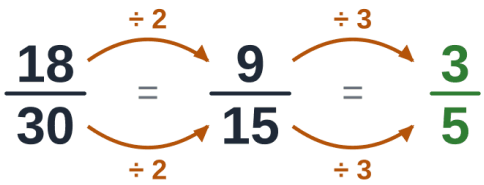
Why? 9 ÷ 3 = 3 and 15 ÷ 3 = 5.

$$\frac{9}{15} = \frac{9 \div 3}{15 \div 3} = \frac{3}{5}$$

STEP 3

Can 3/5 be simplified again? No: 1 is the only number that goes into both 3 and 5. Finished.

Why? Small steps are safe steps. You do not have to find the biggest number in one go.



$$\frac{18}{30} = \frac{3}{5}$$

ANSWER

$$\frac{3}{5}$$

Check: Go back: 3 × 6 = 18 and 5 × 6 = 30 ✓

QUICK CHECK

Your go: simplify 10/15

Which number goes into both 10 and 15? Type the simplified fraction.

$$\frac{10}{15}$$

ANSWER

$$\frac{10}{15} = \frac{10 \div 5}{15 \div 5} = \frac{2}{3}$$

Both end in 0 or 5, so 5 goes into both: 10 ÷ 5 = 2 and 15 ÷ 5 = 3. Then 1 is the only number that goes into both 2 and 3.

Part 3 of 7: Improper fractions and mixed numbers

More slices than one whole holds

PICTURE IT

When the top is bigger than the bottom

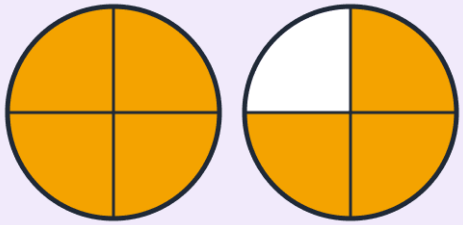
7/4 has more slices than one pizza holds. A fraction like that is called an **improper fraction**.

Divide the top by the bottom: how many 4s fit into 7? One 4 fits, and 3 slices are left over.

So 7/4 is 1 whole pizza and 3 quarters: **1 3/4**. That is a **mixed number**.

7 ÷ 4 = 1 remainder **3**

$$\frac{7}{4} = 1\frac{3}{4}$$



Worked example 2: improper fraction → mixed number

Example 2

Write as a mixed number:

$$\frac{11}{4}$$

STEP 1

Divide the top by the bottom: how many 4s fit into 11? Two 4s make 8. Three 4s make 12: too many. So 2 fit, and 11 − 8 = 3 are left over.

Why? Every 4 quarters make one whole.

11 ÷ 4 = 2 remainder **3**

STEP 2

The 2 is the whole number. The 3 left over is the new top. The slices are still quarters, so the bottom stays 4.

Why? The picture shows 2 whole pizzas and 3 quarters more.



$$\frac{11}{4} = 2\frac{3}{4}$$

ANSWER

$$2\frac{3}{4}$$

Check: Go back: 2 wholes are 8 quarters, plus 3 more makes 11 quarters ✓

TOOL

And back again: mixed number → improper fraction

Multiply the whole number by the bottom. That turns the wholes into slices.

Add the top. Those are the extra slices.

Write the total over the **same bottom**.

$$2\frac{1}{3} = \frac{2 \times 3 + 1}{3} = \frac{7}{3}$$



3 + 3 + 1 = 7 thirds

QUICK CHECK

Fill in the blanks: 2 1/2 as an improper fraction

Type only the two missing numbers, in order.

$$2\frac{1}{2} = \frac{2 \times ? + 1}{2} = \frac{?}{2}$$

ANSWER

$$2\frac{1}{2} = \frac{2 \times 2 + 1}{2} = \frac{5}{2}$$

Whole number × bottom: 2 × 2 = 4 halves. Plus the 1 half on top: 5 halves.

Part 4 of 7: The sign × means "groups of": a whole number times a fraction

Multiply the top only

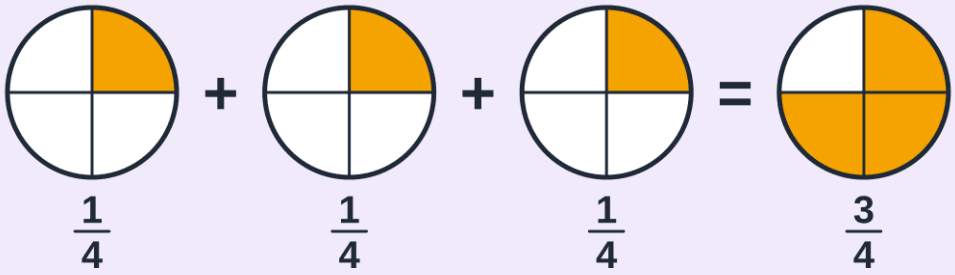
PICTURE IT

3 × 1/4 means 3 groups of a quarter

Take a quarter of a pizza, three times.

Count the slices: 1 quarter + 1 quarter + 1 quarter = **3 quarters**.

$$3 \times \frac{1}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$



THE RULE

A whole number times a fraction: multiply the TOP only

The **top** counts the slices. 3 groups of 1 slice is 3 slices. So **multiply the top** by the whole number.

The **bottom** says which slices they are: quarters. They are still quarters. So **the bottom stays the same**.

$$3 \times \frac{1}{4} = \frac{3 \times 1}{4} = \frac{3}{4}$$

CAREFUL

Do NOT multiply the bottom as well

If you multiply the top AND the bottom by 3, you get 3/12.

3/12 simplifies back to 1/4. The amount has not changed at all. (That is the golden rule.)

Three groups of a quarter must be MORE than one quarter.

Top and bottom multiplied ✗

$$3 \times \frac{1}{4} = \frac{3 \times 1}{4 \times 3} = \frac{3}{12}$$

Only the top multiplied ✓

$$3 \times \frac{1}{4} = \frac{3 \times 1}{4} = \frac{3}{4}$$

Worked example 3: 6 groups of two thirds (Lesson 3, Example 1)

Example 3 · Lesson 3, Worked example 1

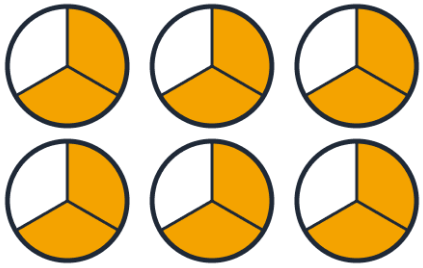
Work out:

$$6 \times \frac{2}{3}$$

STEP 1

Read the question out loud: "6 times two thirds". It means 6 groups of two thirds of a pizza.

Why? Saying it in words first tells you what to expect: many slices.



STEP 2

Multiply the TOP by the whole number: 6 × 2 = 12. That is how many slices there are.

Why? Each group has 2 slices and there are 6 groups: 2 + 2 + 2 + 2 + 2 + 2 = 12.

$$6 \times 2 = 12$$

STEP 3

The bottom stays 3, because every slice is still a third. Write 12 over 3.

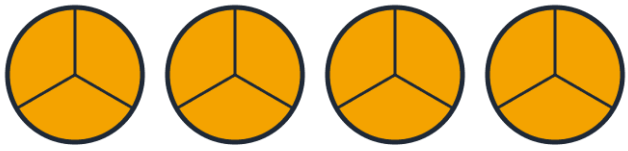
Why? Multiplying changes HOW MANY slices you have. It does not change the size of the slices.

$$6 \times \frac{2}{3} = \frac{6 \times 2}{3} = \frac{12}{3}$$

STEP 4

The top (12) is bigger than the bottom (3), so there are whole pizzas inside. Divide the top by the bottom: how many 3s fit into 12? Four 3s make 12 exactly. Nothing is left over.

Why? Every 3 thirds make one whole pizza. 12 thirds make 4 whole pizzas.



12 thirds = **4** whole pizzas

$$12 \div 3 = 4$$

$$\frac{12}{3} = 4$$

ANSWER

4

Check: Two thirds is a little more than half a pizza. Six halves would be 3 pizzas. So the answer should be a little more than 3 ✓

QUICK CHECK

Fill in the blanks: 4 × 2/5

Type only the three missing numbers, in order.

$$4 \times \frac{2}{5} = \frac{? \times 2}{5} = \frac{?}{5} = 1\frac{?}{5}$$

ANSWER

$$4 \times \frac{2}{5} = \frac{4 \times 2}{5} = \frac{8}{5} = 1\frac{3}{5}$$

Multiply the top by the whole number: 4 × 2 = 8. The bottom stays 5. Then 8 ÷ 5 = 1 remainder 3: one whole and 3 fifths.

Part 5 of 7: A fraction OF something: "of" means ×

Fraction × fraction: top × top, bottom × bottom, then simplify

PICTURE IT

What if the number in front is a fraction? Half a group

2 × 6 is **two groups** of 6. That is 12.

1 × 6 is **one group** of 6. That is 6.

1/2 × 6 is **half a group** of 6. That is 3.

So **1/2 × 6** means **"half OF 6"**. In maths, the word **"of"** means **×**.



PICTURE IT

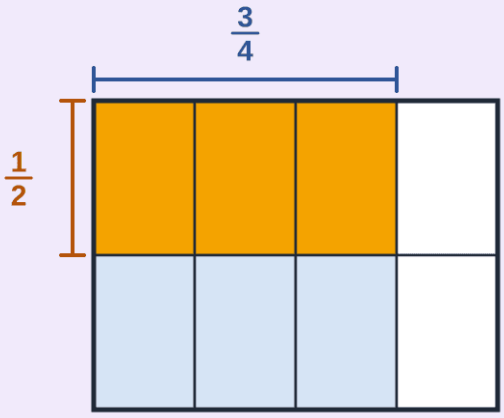
Half OF three quarters of a slab of chocolate

The slab has 4 columns. **3/4** of it is yours: the 3 coloured columns.

You give away **half OF your part**. Cut across the middle and take the top row: the orange part.

The whole slab is now in 8 small pieces. The orange part is **3 of those 8**.

$$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$



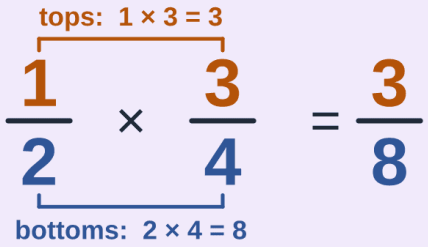
3 of the 8 small pieces are orange

WHY IT WORKS

Where do the 3 and the 8 come from?

The 8 on the bottom: 2 rows × 4 columns = 8 small pieces in the whole slab. That is **bottom × bottom**.

The 3 on top: 1 row × 3 columns = 3 orange pieces. That is **top × top**.



THE RULE

To multiply fractions: top × top, bottom × bottom, then simplify

1. Multiply the **tops**. That is the new top.

2. Multiply the **bottoms**. That is the new bottom.

3. Simplify the answer if you can.



Worked example 4: two thirds of five sixths (Lesson 3, Example 2)

Example 4 · Lesson 3, Worked example 2 (module guide Example 1.6, page 27)

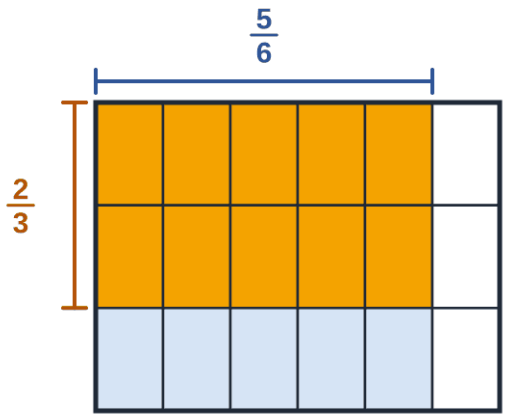
Work out:

$$\frac{2}{3} \times \frac{5}{6}$$

STEP 1

Read the question: "two thirds TIMES five sixths". That means two thirds OF five sixths. In the slab picture: 5 of the 6 columns, then 2 of the 3 rows.

Why? The orange part is the answer. It must be smaller than five sixths, because we only take part of it.

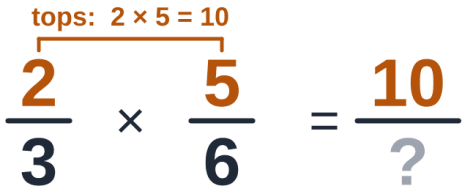


10 of the 18 small pieces are orange

STEP 2

Tops first: 2 × 5 = 10. Write 10 on top of the answer.

Why? In the slab picture: 2 rows × 5 columns = 10 orange pieces.

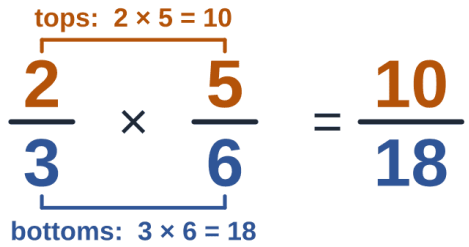


2 × 5 = 10

STEP 3

Now the bottoms: 3 × 6 = 18. Write 18 underneath.

Why? In the slab picture: 3 rows × 6 columns = 18 small pieces in the whole slab.



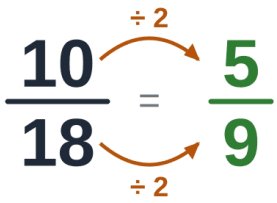
$$3 \times 6 = 18$$

$$\frac{2}{3} \times \frac{5}{6} = \frac{2 \times 5}{3 \times 6} = \frac{10}{18}$$

STEP 4

Last, simplify 10/18. Both are even, so 2 goes into both. Divide the top and the bottom by 2.

Why? Half of 10 is 5. Half of 18 is 9. Then 1 is the only number that goes into both 5 and 9, so it is finished.



$$\frac{10}{18} = \frac{10 \div 2}{18 \div 2} = \frac{5}{9}$$

ANSWER

$$\frac{5}{9}$$

Check: Count the slab picture: 10 of the 18 small pieces are orange, and 10/18 simplifies to 5/9 ✓

QUICK CHECK

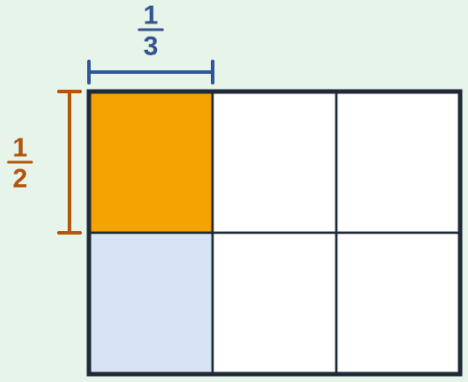
Your go: 1/2 × 1/3

Half OF a third. Top × top, bottom × bottom.

$$\frac{1}{2} \times \frac{1}{3}$$

ANSWER

$$\frac{1}{2} \times \frac{1}{3} = \frac{1 \times 1}{2 \times 3} = \frac{1}{6}$$



1 of the 6 small pieces is orange

Tops: 1 × 1 = 1. Bottoms: 2 × 3 = 6. Half of a third is a sixth. Nothing to simplify.

Part 6 of 7: A whole number or a mixed number in a multiplication

Turn everything into a plain fraction first

THE RULE

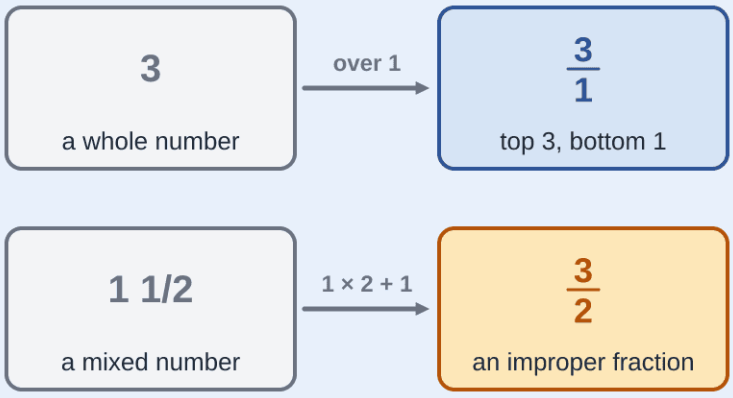
First turn every number into a plain fraction

A **plain fraction** is just a top and a bottom, with no whole number in front.

A whole number: write it over 1. 3 = 3/1, because 3 ÷ 1 is still 3.

A mixed number: change it into an improper fraction (Part 3).

Then multiply as before: top × top, bottom × bottom.



Worked example 5: three groups of one and a half

Example 5 · Lesson 3, "Your go" at the end of Idea 3

Work out:

$$3 \times 1\frac{1}{2}$$

STEP 1

Make both numbers plain fractions. 3 is 3/1. And 1 1/2 is 3/2, because 1 × 2 + 1 = 3.

Why? One whole is 2 halves, plus 1 more half makes 3 halves.

$$3 = \frac{3}{1}$$

$$1\frac{1}{2} = \frac{1 \times 2 + 1}{2} = \frac{3}{2}$$

STEP 2

Tops: 3 × 3 = 9. Bottoms: 1 × 2 = 2.

Why? Top × top, bottom × bottom.

$$\frac{3}{1} \times \frac{3}{2} = \frac{3 \times 3}{1 \times 2} = \frac{9}{2}$$

STEP 3

The top is bigger than the bottom. Divide: how many 2s fit into 9? Four 2s make 8, and 1 is left over.

Why? Every 2 halves make one whole.

$$9 \div 2 = 4 \text{ remainder } 1$$

$$\frac{9}{2} = 4\frac{1}{2}$$

ANSWER

$$4\frac{1}{2}$$

Check: In money: 3 groups of R1.50 is R4.50 ✓

Part 7 of 7: A fraction OF an amount

Divide by the bottom, then multiply by the top

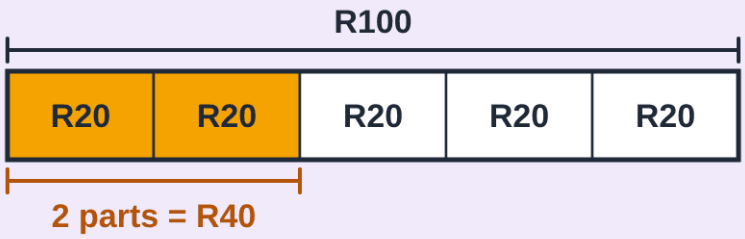
PICTURE IT

What is 2/5 of R100?

"2/5 OF R100" means: share the R100 into **5 equal parts**, then take **2 of the parts**.

One part: R100 ÷ 5 = R20. (Five R20 notes make R100.)

Two parts: 2 × R20 = **R40**.



THE RULE

Divide by the bottom, then multiply by the top

÷ bottom gives the size of ONE part. (The bottom says how many equal parts there are.)

× top takes as many parts as you need. (The top says how many parts you want.)

100 ÷ 5 = 20

2 × 20 = 40

TOOL

Dividing a number that ends in 0

For **120 ÷ 4**, leave the 0 off: 12 ÷ 4 = 3. Then put the 0 back: **30**.

Why? 120 is 12 tens. 12 tens shared by 4 is 3 tens, which is 30.

Check by multiplying back: 4 × 30 = 120 ✓

Worked example 6: the staff canteen (Lesson 3, Example 5)

Example 6 · Lesson 3, Worked example 5

A company has 120 staff. 3/4 of them use the canteen. How many staff use the canteen?

STEP 1

Find the fraction and the amount. The fraction is 3/4. The amount is 120 staff. "3/4 OF 120" means: split the staff into 4 equal groups and take 3 of the groups.

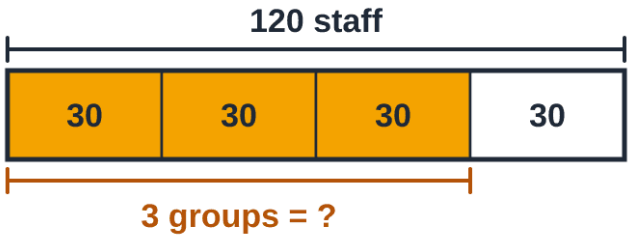
Why? The bottom (4) says how many equal groups. The top (3) says how many groups we want.



STEP 2

Divide by the bottom to find ONE group: 120 ÷ 4. Think of 12 ÷ 4 = 3, so 120 ÷ 4 = 30.

Why? 120 is 12 tens. 12 tens shared by 4 is 3 tens, which is 30. Check: 30 + 30 + 30 + 30 = 120.

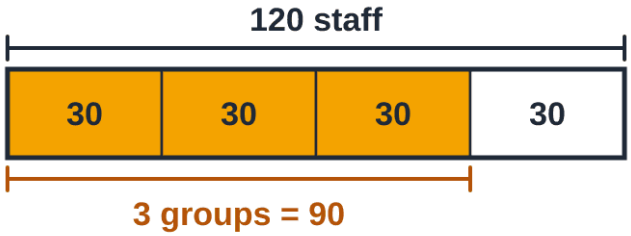


$$120 \div 4 = 30$$

STEP 3

Multiply by the top to get THREE groups: 3 × 30 = 90.

Why? 3 groups of 30: 30 + 30 + 30 = 90.



$$3 \times 30 = 90$$

STEP 4

Check: the one group left over (30 people) does not use the canteen. The two numbers together must be everyone.

Why? The people who use it plus the people who do not must add up to all 120 staff.

$$90 + 30 = 120$$

ANSWER

90 staff use the canteen.

LOOK

You have caught up with the group

The group stopped at the end of this example.

Next session you continue together from Lesson 3, Worked example 6 ("the part that is left").

After that comes Idea 5 of Lesson 3: dividing fractions.

Homework

Do these before the next session. Cover the working, try the question, then compare line by line. Put a star next to any step you did not understand.

Homework 1

Simplify:

$$\frac{16}{24}$$

STEP 1

Both are even: divide the top and the bottom by 2.

$$\frac{16}{24} = \frac{16 \div 2}{24 \div 2} = \frac{8}{12}$$

STEP 2

Both are still even: divide by 2 again.

$$\frac{8}{12} = \frac{8 \div 2}{12 \div 2} = \frac{4}{6}$$

STEP 3

Still even: divide by 2 once more. Then 1 is the only number that goes into both 2 and 3. Finished.

$$\frac{4}{6} = \frac{4 \div 2}{6 \div 2} = \frac{2}{3}$$

ANSWER

$$\frac{2}{3}$$

Check: Go back: 2 × 8 = 16 and 3 × 8 = 24 ✓

Homework 2

Write as a mixed number:

$$\frac{13}{4}$$

STEP 1

How many 4s fit into 13? Three 4s make 12. Four 4s make 16: too many. So 3 fit, and 13 − 12 = 1 is left over.

$$13 \div 4 = 3 \text{ remainder } 1$$

STEP 2

The 3 is the whole number. The 1 left over is the new top. The bottom stays 4.

$$\frac{13}{4} = 3\frac{1}{4}$$

ANSWER

$$3\frac{1}{4}$$

Homework 3

Work out:

$$4 \times \frac{3}{8}$$

STEP 1

A whole number times a fraction: multiply the top only. 4 × 3 = 12. The bottom stays 8.

$$4 \times \frac{3}{8} = \frac{4 \times 3}{8} = \frac{12}{8}$$

STEP 2

Simplify: 12 and 8 are both in the 4 times table. Divide both by 4.

$$\frac{12}{8} = \frac{12 \div 4}{8 \div 4} = \frac{3}{2}$$

STEP 3

The top is bigger than the bottom: 3 ÷ 2 = 1 remainder 1.

$$\frac{3}{2} = 1\frac{1}{2}$$

ANSWER

$$1\frac{1}{2}$$

Homework 4

Work out:

$$\frac{2}{5} \times \frac{3}{4}$$

STEP 1

Top × top, bottom × bottom.

$$\frac{2}{5} \times \frac{3}{4} = \frac{2 \times 3}{5 \times 4} = \frac{6}{20}$$

STEP 2

Simplify: both are even, so divide both by 2. Then 1 is the only number that goes into both 3 and 10.

$$\frac{6}{20} = \frac{6 \div 2}{20 \div 2} = \frac{3}{10}$$

ANSWER

$$\frac{3}{10}$$

Homework 5

A shop takes R240 in a morning. 3/8 of it is for cold drinks. How much is that?

STEP 1

Divide by the bottom to find ONE part: 240 ÷ 8. Leave the 0 off: 24 ÷ 8 = 3. Put the 0 back: 30.

Why? Row 8 of the times table: 8 × 3 = 24.

$$240 \div 8 = 30$$

STEP 2

Multiply by the top to get THREE parts.

$$3 \times 30 = 90$$

ANSWER

R90 is for cold drinks.

Check: Rough check: 3/8 is a little less than half. Half of R240 is R120, and R90 is a little less ✓

Homework 6: reading list

Read these parts of your notes, in this order, before the BODMAS lesson. Try each question before you look at its working.

STEP 1

Lesson 2 notes, Ideas 4 to 6: adding and subtracting fractions. Lesson 3 uses them again in its homework, and the BODMAS lesson needs them. In Idea 5, find the LCD by listing multiples. You can skip the prime ladder: the group does not use it any more.

STEP 2

Lesson 1 in the Student Workbook, Ideas 1 and 2: place value, rounding and negative numbers. The decimals lesson needs the rounding. The BODMAS lesson needs the negative numbers.

STEP 3

Stuck on something? Put a star next to it and bring it to the next session.

ANSWER

No answer to check: the notes have every question worked in full.

Today in a few lines

1. **×** means "groups of". **÷** asks "how many groups fit?" The times table does both.
2. A fraction: the **bottom** says how many equal slices make one whole. The **top** counts the slices you have.
3. **Simplify:** divide the top and the bottom by a number that goes into both, until only 1 goes into both.
4. **Improper → mixed:** divide the top by the bottom. **Mixed → improper:** whole number × bottom, plus the top.
5. **Whole number × fraction:** multiply the top only. **Fraction × fraction:** top × top, bottom × bottom, then simplify.
6. **A fraction OF an amount:** divide by the bottom, then multiply by the top.

Next lesson: With the group: Lesson 3 from Worked example 6 ("the part that is left"), then dividing fractions.